BUSFIN 8200 – Problem Set 2

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October 26, 2026

Question 1

Question 2

2 (a)

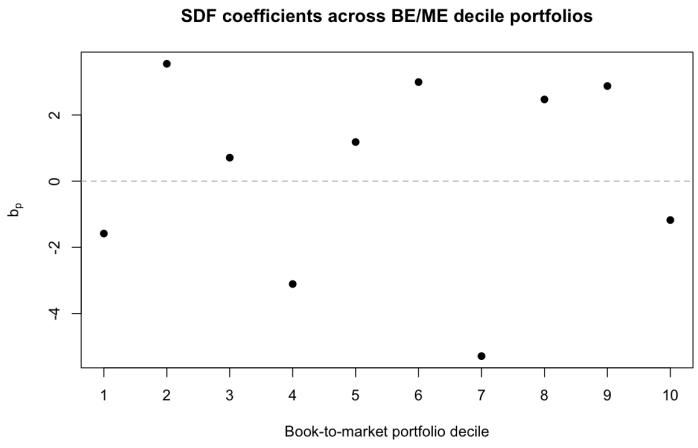
Using the normalization $\mathbb{E}[M^*] = 1$, we find the stochastic discount factor $M_t^* = a - b'xR_t$ that satisfies $\hat{\mathbb{E}}[M^* \cdot xR] = 0$. In particular, we have:

$$\begin{aligned} 0 &= \mathbb{E}[M^* \cdot xR] \\ &= \underbrace{\mathbb{E}[M^*]}_{=1} \cdot \mathbb{E}[xR] + \text{Cov}(M^*, xR) \\ &= \mathbb{E}[xR] - b'\text{Cov}(xR, xR) \\ &\implies \mathbb{E}[xR] = b'\Sigma \\ &\implies b = \Sigma^{-1}\mathbb{E}[xR] \end{aligned}$$

xR is a Tx10 vector of excess returns with T = 1,202 months in the sample from July 1926 to August 2026. Verifying that Σ is invertible, we find that:

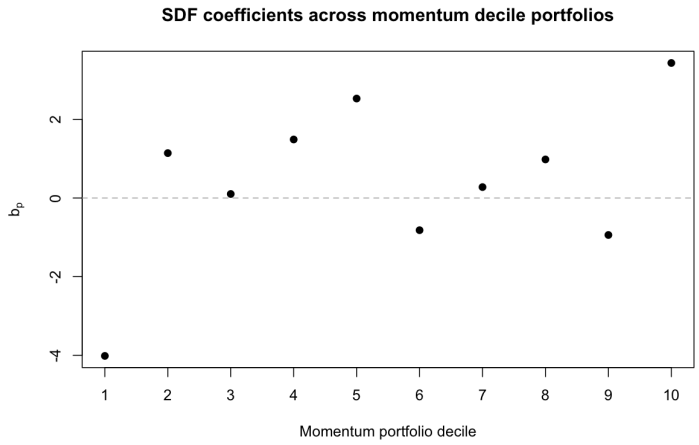
$$\begin{aligned} 1 &= \mathbb{E}[M^*] \\ &= \mathbb{E}[a - b'xR] \\ &= a - b'\mathbb{E}[xR] \\ &\implies a = 1 + b'\mathbb{E}[xR] \end{aligned}$$

Therefore, we are able to compute the stochastic discount factor $M_t^* = a - b'xR_t$ for each month in the sample and obtain the following scatterplot:



2 (b)

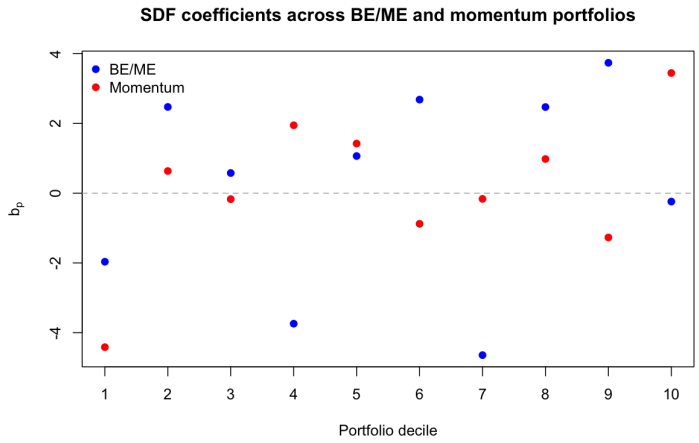
Following the same procedure as in part (a), we find the respective b coefficients of the stochastic discount factor that prices the excess returns of the 10 portfolios in the MOM dataset as follows:



Similar to part (a), we do not observe any clear pattern between the b coefficients and the decile portfolios' excess returns. The correlation between the recovered M_t^* is 0.3555, which suggests that the SDF that prices a particular portfolio does not necessarily price the other one as the decile portfolios are sorted using different signals.

2 (c)

Following the same procedure as in part (a), we find the respective b coefficients of the stochastic discount factor that prices the excess returns of the 20 portfolios in the BM & MOM datasets as follows:



Using a common sample of 1,196 months from July 1927 to August 2026, we estimated the Sharpe ratios of xR_t^* as follows:

Table 1: Monthly Sharpe Ratios of the xR^* Portfolios

Portfolio set	Monthly Sharpe ratio
BE/ME (10 portfolios)	0.191841
Momentum (10 portfolios)	0.257990
Combined (20 portfolios)	0.304398

A key property of the SDF under no arbitrage is that xR_t^* is the maximum Sharpe ratio portfolio. Since we observed that the Sharpe ratio of the combined 20 portfolios is higher than the Sharpe ratios of the individual 10 portfolios, adding more non-redundant assets to the portfolio universe provides a more diversified portfolio with better risk-adjusted returns and higher volatility, which is consistent with the theory of SDF and maximum Sharpe ratio portfolios.

2 (d)

Having estimated the SDFs $M_t^* = \hat{a} - \hat{b}' xR_t$ from part (a),(b), and (c) and normalizing $\mathbb{E}[M^*] = 1$, we estimate the pricing errors for each decile portfolio p as follows:

$$\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] + \widehat{\mathrm{Cov}}(M^*, xR_p).$$

Table 2: Pricing Errors Under the Three Estimated SDFs

Portfolio	BE/ME SDF	Momentum SDF	Combined SDF
BE/ME D1	0.000005	-0.001505	0.000005
BE/ME D2	0.000006	-0.000095	0.000006
BE/ME D3	0.000006	-0.000217	0.000006
BE/ME D4	0.000006	-0.000772	0.000006
BE/ME D5	0.000006	0.000509	0.000006
BE/ME D6	0.000007	0.001267	0.000007
BE/ME D7	0.000006	0.000287	0.000006
BE/ME D8	0.000008	0.002382	0.000008
BE/ME D9	0.000009	0.003456	0.000009
BE/ME D10	0.000009	0.003204	0.000009
Momentum D1	-0.010147	0.000001	0.000001
Momentum D2	-0.005043	0.000004	0.000004
Momentum D3	-0.003265	0.000004	0.000004
Momentum D4	-0.001482	0.000005	0.000005
Momentum D5	-0.000875	0.000006	0.000006
Momentum D6	-0.000257	0.000006	0.000006
Momentum D7	0.000515	0.000006	0.000006
Momentum D8	0.001908	0.000007	0.000007
Momentum D9	0.002246	0.000008	0.000008
Momentum D10	0.005424	0.000010	0.000010

We observe that the SDF constructed using the BE/ME portfolios prices the BE/ME portfolios well with small pricing errors, but not the MOM portfolios, and vice versa. The SDF based on the combined 20 portfolios prices all 20 portfolios well, which is consistent with the notion that the selection of the asset universe matters in the pricing power of the SDF.

2 (e)

Repeating the same procedure as in part (c) but restricting the sample to December 1989, I re-estimate $\hat{a}_{OS} = 1.12459$ and corresponding $\hat{b}_{OS}$ coefficients for the SDF that prices the 20 portfolios in the BM & MOM datasets. Then, using out-of-sample xR_t from January 1990 to August 2026, I construct $M^*_{OS,t} = \hat{a}_{OS} - \hat{b}'_{OS} xR_t$ and obtain the following pricing errors:

Table 3: Out-of-Sample Pricing Errors

Portfolio	OS pricing error
BE/ME D1	0.010968
BE/ME D2	0.007859
BE/ME D3	0.009001
BE/ME D4	0.006958
BE/ME D5	0.007985
BE/ME D6	0.007179
BE/ME D7	0.006975
BE/ME D8	0.005926
BE/ME D9	0.008271
BE/ME D10	0.009709
Momentum D1	0.027023
Momentum D2	0.016469
Momentum D3	0.013896
Momentum D4	0.012071
Momentum D5	0.009178
Momentum D6	0.008297
Momentum D7	0.007601
Momentum D8	0.006446
Momentum D9	0.004326
Momentum D10	0.005462

Contrary to the in-sample results, the out-of-sample pricing errors are much larger, even for the combined SDF, which suggests the possibility of time-varying a and b coefficients in the SDF.

Question 3

3 (a)

Consider a representative investor with wealth W_t in a two-period economy. Let $R_{f,t}$ be a baseline asset. We solve the following optimization problem:

$$\varpi^*_t = \arg\max_{\varpi_t} \mathbb{E}_t[U(W_{t+1})]\,,$$
$$\text{s.t.} \quad W_{t+1} = W_t \cdot R_{w,t+1} = W_t \cdot (R_{f,t+1} + \varpi'_t xR_{t+1}),$$

where xR_{t+1} is the vector of asset excess returns relative to $R_{f,t}$.
Using the power utility function

$$U(W) = \frac{W^{1-\gamma}}{1-\gamma},$$

where γ is CRRA, we rewrite the problem as

$$\varpi^*_t = \arg\max_{\varpi_t} \mathbb{E}_t \left[\frac{(W_t \cdot (R_{f,t+1} + \varpi'_t xR_{t+1}))^{1-\gamma}}{1-\gamma} \right].$$

Thus, the first-order condition of the maximization problem is

$$\begin{aligned}\mathbb{E}_t\left[\frac{(1-\gamma)[W_t\cdot(R_{f,t+1}+\varpi_t' xR_{t+1})]^{-\gamma}\cdot xR_{t+1}}{1-\gamma}\right]&=0\\ \implies \mathbb{E}_t\Big[\big(R_{f,t+1}+\varpi_t' xR_{t+1}\big)^{-\gamma}\cdot xR_{t+1}\Big]&=0\\ \implies \mathbb{E}_t\Big[R_{m,t+1}^{-\gamma}\cdot xR_{t+1}\Big]&=0.\end{aligned}$$

As a result, the SDF $M_t = R_{m,t+1}^{-\gamma}$ prices the vector of excess returns.

3 (b)

The moment function is given by $\epsilon(\gamma) = R_{w,t+1}^{-\gamma}\cdot xR_{t+1}$. Its sample estimator is $g_T(\gamma) = \frac{1}{T}\sum_{t=1}^{T}\epsilon(\gamma)$. Since there is 1 moment condition and 1 parameter, the GMM estimator $\hat{\gamma}$ is obtained by solving the minimization problem $\hat{\gamma} = \arg\min_\gamma g_T(\gamma)' g_T(\gamma)$, where $W = I$. Using the same sample of 1,129 months from December 1927 to December 2021, we obtain the following GMM estimate of risk aversion:

Table 4: Single-moment GMM Estimate of Risk Aversion

Statistic	Value
$\hat{\gamma}$	1.9061
Newey–West standard error	0.6026
$\hat{\gamma}$ t-statistic	3.1630
$\hat{G}$	−0.0480
$\hat{S}^{\text{HAC}}$	0.9439

Note: The Newey–West bandwidth is 25.5583. Ceiling rounding gives pilot truncation $m = 7$ and $L = 26$ Bartlett lags. Prewhitening is not used.

3 (c)

Following the same procedure in Question 2 part (e) in Problem Set 1, I estimate the out-of-sample forecast of $xR_{e,t+1}$, $\hat{\mathbb{E}}_t^{OS}[xR_e]$, and construct a managed portfolio using $z_t = \hat{\mathbb{E}}_t^{OS}[xR_e]/\sigma_e^2$ as a trading signal based on the dividend-price ratio. Thus, given 1 parameter γ, we have 2 moment conditions:

$$\begin{aligned}\mathbb{E}\Big[R_{w,t}^{-\gamma}\cdot xR_{w,t}\Big]&=0,\\ \mathbb{E}\Big[R_{w,t}^{-\gamma}\cdot(z_{t-1}\cdot xR_{w,t})\Big]&=0.\end{aligned}$$

Using an identity weighting matrix, we have the following optimization problem:

$$\hat{\gamma} = \arg\min_{\gamma} g_T(\gamma)' g_T(\gamma),$$

where $g_T(\gamma) = \frac{1}{T}\sum_{t=1}^{T}\epsilon(\gamma)$ and $\epsilon(\gamma) = \begin{bmatrix} R_{w,t}^{-\gamma}\cdot xR_{w,t} \\ R_{w,t}^{-\gamma}\cdot(z_{t-1}\cdot xR_{w,t})\end{bmatrix}$.
Solving for $\hat{\gamma}$ using the same sample of 973 months from December 1940 to December 2021, we obtain the following GMM estimate of risk aversion:

Table 5: Identity-Weighted GMM with Market and Managed-Portfolio Moments

Panel A: Parameter and Pricing Errors

Parameter / Moment	Estimate	NW SE	t-statistic
$\hat{\gamma}$	3.0064	0.7207	4.1718
Market pricing error	−0.0038	0.0048	−0.7934
Managed pricing error	0.0019	0.0023	0.7934

Panel B: Joint Wald Test

Null hypothesis	Wald statistic	df	p-value
$\alpha_{\text{market}} = \alpha_{\text{managed}} = 0$	0.6294	1	0.4276

Note: The sample covers December 1940–December 2021 ($T = 973$). NW SE denotes the Newey–West standard error. The joint test uses the pseudoinverse of the pricing-error covariance matrix, which has numerical rank one. The Bartlett bandwidth is 23.3785; ceiling rounding gives pilot truncation $m = 7$ and $L = 24$ lags. Lag-ℓ autocovariances use denominator $T - \ell$; prewhitening is not used. The optimum is interior. The objective grid with spacing 0.01 and sensitivity checks over $[-20, 20]$ and $[-50, 50]$ yield estimates agreeing to four decimal places.

The p-value of the joint Wald test is 0.4276, which suggests that we fail to reject the null hypothesis that the estimated pricing errors are jointly equal to zero. Thus, the estimated SDF is consistent in pricing both the market and managed portfolios.

3 (d)

I repeat the same procedure in part (c) as the first step and use $W = \hat{S}_{\text{HAC}}^{-1}$ as the optimal weighting matrix in the second step.

Table 6: Two-Step GMM with Market and Managed-Portfolio Moments

Panel A: Parameter and Pricing Errors

Parameter / Moment	Estimate	NW SE	t-statistic
$\hat{\gamma}_2$	2.9508	0.7106	4.1524
Market pricing error	−0.0020	0.0025	−0.8078
Managed pricing error	0.0055	0.0068	0.8078

Panel B: Joint Wald Test

Null hypothesis	Wald statistic	df	p-value
$\alpha_{\text{market}} = \alpha_{\text{managed}} = 0$	0.6525	1	0.4192

Note: The sample covers December 1940–December 2021 ($T = 973$). NW SE denotes the Newey–West standard error. The joint test uses the pseudoinverse of the pricing-error covariance matrix, which has numerical rank one. Stage-one and stage-two Bartlett bandwidths are 23.3785 and 23.4192, respectively, giving $L_1 = L_2 = 24$ after ceiling rounding. Lag-ℓ autocovariances use denominator $T - \ell$; prewhitening is not used. The stage-one eigenvalue ratio is 0.0100. The optimum is interior; the objective grid with spacing 0.01 and wider-bound checks yield estimates agreeing to four decimal places.

Again, we fail to reject the null hypothesis that the estimated pricing errors are jointly equal to zero, which suggests that the estimated SDF is consistent in pricing both the market and managed portfolios.

3 (e)

Since the BM and MOM portfolio returns and risk-free rates are monthly observations, I convert percentage returns to decimals andcalculate the annual gross return over 12 months ending at t for each decile portfolio p as follows:

$$R_{p,t}^{\text{annual}} = \prod_{k=0}^{11}(1 + R_{p,t-k}).$$

Then, I calculate the annual excess return of each decile portfolio p relative to the annuallycompounded risk-free rate. I estimate the GMM using 21 pricing moments: 1 for xR_w, 10 for the BM decile portfolios, and 10 for the MOM decile portfolios. The moment function is given by $\epsilon(\gamma) = R_{w,t+1}^{-\gamma} \cdot xR_{t+1}$, where xR_{t+1} is a Tx21 vector of excess returns with T = 1,129 months. Following the same GMM procedure in part 9=(c), I obtain the following GMM estimate of risk aversion:

Table 7: Identity-Weighted GMM with 21 Pricing Moments

Parameter / Moment	Estimate	NW SE	t-statistic
$\hat{\gamma}$	1.9868	0.5910	3.3619
Market pricing error	-0.0039	0.0039	-1.0018
BM D1 pricing error	-0.0092	0.0103	-0.8927
BM D2 pricing error	0.0103	0.0072	1.4312
BM D3 pricing error	0.0024	0.0062	0.3929
BM D4 pricing error	-0.0017	0.0058	-0.2982
BM D5 pricing error	0.0083	0.0070	1.1973
BM D6 pricing error	0.0149	0.0082	1.8119
BM D7 pricing error	-0.0009	0.0093	-0.0914
BM D8 pricing error	0.0223	0.0081	2.7555
BM D9 pricing error	0.0330	0.0095	3.4950
BM D10 pricing error	0.0188	0.0146	1.2870
MOM D1 pricing error	-0.1018	0.0128	-7.9428
MOM D2 pricing error	-0.0439	0.0086	-5.1308
MOM D3 pricing error	-0.0301	0.0094	-3.2136
MOM D4 pricing error	-0.0117	0.0071	-1.6506
MOM D5 pricing error	-0.0012	0.0068	-0.1836
MOM D6 pricing error	-0.0004	0.0048	-0.0766
MOM D7 pricing error	0.0112	0.0070	1.6122
MOM D8 pricing error	0.0249	0.0057	4.3830
MOM D9 pricing error	0.0302	0.0081	3.7248
MOM D10 pricing error	0.0620	0.0130	4.7810
Joint pricing-error test	201.5580	$df = 20$	$p = 0.0000$

Note: The sample covers December 1927–December 2021 ($T = 1129$). The Bartlett bandwidth is 25.3665; ceiling rounding gives pilot truncation $m = 7$ and $L = 26$. Lag-ℓ autocovariances use denominator $T - \ell$ and no prewhitening. The pricing-error covariance has numerical rank 20 under relative tolerance 10^{-10}. The optimizer is interior; the 0.01 grid and wider-bound checks passed.

We observe that the p-value of the joint pricing-error test is less than 0.05, which suggests that we reject the null hypothesis that the estimated pricing errors are jointly equal to zero. As a result, the estimated SDF is not consistent in pricing the market and 20 decile portfolios.

Question 4

4 (a)

The SDF = $M_t = R_{w,t+1}^{-\gamma}$ is non-linear in γ and can be written as $f(R_{w,t+1}, \gamma)$. Using first-order Taylor approximation, which gives $M_{t+1} = f(F_{t+1}, \theta) \approx f(\mathbb{E}_t[F], \theta) + \partial_F f(\mathbb{E}_t[F], \theta)'(F_{t+1} - \mathbb{E}_t[F])$, we can linearize the SDF as follows:

$$
\begin{aligned}
M_{t+1} &= f(R_{w,t+1}, \gamma) \\
&\approx f(\mathbb{E}_t[R_{w,t+1}], \gamma) + \frac{\partial f}{\partial R}(\mathbb{E}_t[R_{w,t+1}], \gamma)\left(R_{w,t+1} - \mathbb{E}_t[R_{w,t+1}]\right) \equiv M_{t+1}^{\text{Linear}} \\
&= \left(\mathbb{E}_t[R_{w,t+1}]\right)^{-\gamma} - \gamma\left(\mathbb{E}_t[R_{w,t+1}]\right)^{-\gamma-1}\left(R_{w,t+1} - \mathbb{E}_t[R_{w,t+1}]\right) \\
&= \left(\mathbb{E}_t[R_{w,t+1}]\right)^{-\gamma}\left(1 - \gamma\frac{R_{w,t+1} - \mathbb{E}_t[R_{w,t+1}]}{\mathbb{E}_t[R_{w,t+1}]}\right) \\
&= (1+\gamma)\left(\mathbb{E}_t[R_{w,t+1}]\right)^{-\gamma} - \gamma\left(\mathbb{E}_t[R_{w,t+1}]\right)^{-\gamma-1} R_{w,t+1} \\
&= \mathbb{E}_t\left[M_{t+1}^{\text{Linear}}\right]\left((1+\gamma) - \frac{\gamma}{\mathbb{E}_t[R_{w,t+1}]}R_{w,t+1}\right) \\
&= \mathbb{E}_t\left[M_{t+1}^{\text{Linear}}\right](a_t - b_t R_{w,t+1})
\end{aligned}
$$

where

$$
a_t = 1+\gamma, \qquad b_t = -\frac{\partial_R f(\mathbb{E}_t[R_{w,t+1}], \gamma)}{f(\mathbb{E}_t[R_{w,t+1}], \gamma)} = \frac{\gamma}{\mathbb{E}_t[R_{w,t+1}]}, \qquad \mathbb{E}_t\left[M_{t+1}^{\text{Linear}}\right] = \left(\mathbb{E}_t[R_{w,t+1}]\right)^{-\gamma}.
$$

A disadvantage of the linearized SDF is that a_t and b_t are time-varying. In particular, b_t is a function of the conditional expectation of the market return $\mathbb{E}_t[R_{w,t+1}]$, which is not observable. Therefore, errors in estimating $\mathbb{E}_t[R_{w,t+1}]$ can distort b_t, thus affecting the pricing power of the linearized SDF. Indeed, the unconditional implication of the linearized SDF with time-varying parameters is $\mathbb{E}[xR_w] = \text{Cov}(b_t' F_{t+1}, xR_{w,t+1})$, which is not the same as the unconditional SDF coveriance pricing equation $\mathbb{E}[xR_w] = \mathbb{E}[b_t']\,\text{Cov}(F_{t+1}, xR_{w,t+1})$. As a result, we cannot define $b = \mathbb{E}[b_t]$ and treat the model as a linear SDF with constant parameters, which is a key assumption in the GMM estimation procedure.

4 (b)

The SDF is defined as $M_t = a - b \cdot R_{w,t}$, where a and b are constant parameters for simplicity. With the normalization $\mathbb{E}[M_t] = 1$, we have $M_t = 1 - b \cdot (R_{w,t} - \mathbb{E}[R_w])$. Following the same GMM procedure in Question 3 part (e), we obtain the following GMM estimate of b:

Table 8: Linear-SDF Identity-Weighted GMM with 21 Pricing Moments

Parameter / Moment	Estimate	NW SE	t-statistic
$\hat{b}$	1.9955	0.5850	3.4114
Market pricing error	0.0000	0.0046	-0.0058
BM D1 pricing error	-0.0054	0.0116	-0.4644
BM D2 pricing error	0.0163	0.0083	1.9629
BM D3 pricing error	0.0057	0.0069	0.8177
BM D4 pricing error	-0.0019	0.0061	-0.3151
BM D5 pricing error	0.0108	0.0069	1.5597
BM D6 pricing error	0.0162	0.0077	2.0947
BM D7 pricing error	-0.0015	0.0095	-0.1627
BM D8 pricing error	0.0191	0.0088	2.1776
BM D9 pricing error	0.0264	0.0108	2.4385
BM D10 pricing error	0.0079	0.0152	0.5199
MOM D1 pricing error	-0.1113	0.0128	-8.7168
MOM D2 pricing error	-0.0497	0.0088	-5.6619
MOM D3 pricing error	-0.0277	0.0098	-2.8433
MOM D4 pricing error	-0.0130	0.0072	-1.8065
MOM D5 pricing error	0.0001	0.0072	0.0131
MOM D6 pricing error	0.0002	0.0052	0.0366
MOM D7 pricing error	0.0164	0.0073	2.2588
MOM D8 pricing error	0.0297	0.0064	4.6700
MOM D9 pricing error	0.0363	0.0089	4.0869
MOM D10 pricing error	0.0716	0.0139	5.1486
Joint pricing-error test	277.0106	$df = 20$	$p = 0.0000$

Note: The sample covers December 1927–December 2021 ($T = 1129$), with $\overline{R}_w = 1.0877$ treated as fixed in variance calculations. The Bartlett bandwidth is 25.0249; ceiling rounding gives pilot truncation $m = 7$ and $L = 26$. Lag-ℓ autocovariances use denominator $T - \ell$ and no prewhitening. The pricing-error covariance has numerical rank 20 under relative tolerance 10^{-10}. The optimizer is interior; the 0.01 grid and wider-bound checks passed.

4 (c)

Table 9: Efficient Two-Step Linear-SDF GMM with 21 Pricing Moments

Parameter / Moment	Estimate	NW SE	t-statistic
$\hat{b}_2$	4.1665	0.5629	7.4016
Market pricing error	-0.0914	0.0343	-2.6623
BM D1 pricing error	-0.1012	0.0349	-2.8975
BM D2 pricing error	-0.0690	0.0265	-2.6006
BM D3 pricing error	-0.0830	0.0288	-2.8853
BM D4 pricing error	-0.1022	0.0414	-2.4684
BM D5 pricing error	-0.0835	0.0388	-2.1535
BM D6 pricing error	-0.0832	0.0434	-1.9182
BM D7 pricing error	-0.1051	0.0473	-2.2231
BM D8 pricing error	-0.0949	0.0453	-2.0945
BM D9 pricing error	-0.0954	0.0468	-2.0381
BM D10 pricing error	-0.1339	0.0576	-2.3229
MOM D1 pricing error	-0.2464	0.0552	-4.4662
MOM D2 pricing error	-0.1687	0.0511	-3.3020
MOM D3 pricing error	-0.1274	0.0438	-2.9070
MOM D4 pricing error	-0.1161	0.0456	-2.5476
MOM D5 pricing error	-0.0920	0.0362	-2.5374
MOM D6 pricing error	-0.0969	0.0394	-2.4627
MOM D7 pricing error	-0.0718	0.0320	-2.2474
MOM D8 pricing error	-0.0612	0.0296	-2.0652
MOM D9 pricing error	-0.0587	0.0294	-1.9928
MOM D10 pricing error	-0.0300	0.0272	-1.1053
Joint pricing-error test	103.4546	$df = 20$	$p = 0.0000$

Note: The sample covers December 1927–December 2021 ($T = 1129$), with $\overline{R}_w = 1.0877$ treated as fixed. The first- and second-stage Bartlett bandwidths are 25.0249 and 22.5858, giving $L_1 = 26$ and $L_2 = 23$. Lag-ℓ autocovariances use denominator $T - \ell$ and no prewhitening. The first-stage HAC minimum eigenvalue is 0.0020 and its minimum/maximum eigenvalue ratio is 0.0001. The analytical denominator is 0.0067, the weighted first-order condition is -1.2630e-16, and the second-stage pricing-error covariance has numerical rank 20 under relative tolerance 10^{-10}.

4 (d)

4 (e)